BHARATCARES MACHINE LEARNING & APPLIED AI INTERNSHIP

TECHNICAL FINAL PROJECT REPORT

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1. Technical Abstract & Implementation Overview

LearnMate AI was developed to address persistent educational hurdles in engineering education: identifying conceptual deficiencies, maintaining consistent study routines, and accessing high-quality technical tutoring without cost barriers. The implemented solution features a React 19 + TypeScript single-page application communicating with a high-performance Python FastAPI backend. A Scikit-Learn Random Forest Classifier trained on 1,200 synthetic student learning trajectories predicts student support tiers with 88.33% accuracy. An orchestration layer coordinates five specialized local agents to execute diagnostic assessments, curriculum scheduling, concept explanation, and career roadmapping.

2. Repository Architecture & Directory Structure

The project follows a clean decoupled structure separating API endpoints, agent logic, database layer, and ML pipelines:
• backend/app/main.py: FastAPI application entry point with CORS and Lifespan managers.
• backend/app/routes/: Modular route controllers (dashboard, quiz, ml, tutor, planner, progress, agents).
• backend/app/ml/: Synthetic data generator, training script, and runtime Random Forest predictor.
• backend/app/agents/: Master Orchestrator, Tutor Agent, Quiz Agent, Analyst Agent, Planner Agent, Career Agent.
• backend/app/db.py: SQLite schema creation, connection handling, and sample database seeding.
• frontend/src/: React 19 UI components, interactive charts (Recharts), and typed API clients.
• docs/: Required deliverables (Lean Canvas PDF, Concept Note PDF, PowerPoint PPTX, Report PDF).

3. Machine Learning Model Training & Evaluation Results

The supervised classification model predicts three support categories: (0) Needs foundational revision, (1) Developing understanding, and (2) Ready for advanced practice.

- **Training Samples:** 960 (80% split) | **Test Samples:** 240 (20% split) | Random Seed: 42
- **Random Forest Accuracy:** 88.33% | **Macro F1-Score:** 0.8809 | Weighted F1: 0.8812
- **Decision Tree Baseline:** 86.67% accuracy (confirming ensemble performance gain)
- **Primary Contributing Features:** Recent Quiz Average (32.4%), Practice Completion (24.1%), Previous Quiz Score (18.6%).

4. Automated Test Suite Execution Log

Automated regression and integration tests were executed using pytest across all system components. **All 18 test cases passed with 100% success rate:**

✓ test_agents.py::test_tutor_agent_retrieval — Passed
✓ test_agents.py::test_tutor_agent_fallback — Passed
✓ test_agents.py::test_quiz_agent_question_generation — Passed
✓ test_agents.py::test_analyst_agent — Passed
✓ test_agents.py::test_planner_agent — Passed
✓ test_agents.py::test_career_agent — Passed
✓ test_agents.py::test_orchestrator_composite_workflow — Passed
✓ test_api.py::test_health_check — Passed
✓ test_api.py::test_dashboard_summary — Passed
✓ test_api.py::test_quiz_topics_and_submit — Passed
✓ test_api.py::test_ml_prediction_endpoint — Passed
✓ test_api.py::test_ml_model_info — Passed
✓ test_api.py::test_tutor_query — Passed
✓ test_api.py::test_planner_endpoints — Passed
✓ test_api.py::test_agent_orchestrator_execution — Passed
✓ test_ml.py::test_model_loaded_and_predicts — Passed
✓ test_ml.py::test_foundational_revision_prediction — Passed
✓ test_ml.py::test_model_metrics_structure — Passed

5. Ethical Governance, Provenance, and Limitations

• **Dataset Calibration:** Training utilized synthetic student trajectories generated with realistic Gaussian distributions and noise parameters to model student study variations. Provenance is openly disclosed across all deliverables.
• **Educational Safeguards:** Model classifications are strictly utilized to recommend pacing and targeted drills, never to judge human capability or general intelligence.
• **Local Privacy:** Zero telemetry or student answers are dispatched to third-party commercial LLM endpoints.

Submission Sign-Off: LearnMate AI engineering implementation verified and complete.